

HSP90 inhibitors look most promising; WEE1 inhibitors are mechanistically plausible, while senolytics and IGF-1R inhibitors remain less direct.

1. Introduction

For spindle assembly checkpoint deficiency caused by **biallelic BUB1B loss**, the strongest repurposing rationale for targeting **aneuploidy-induced proteotoxic stress** is **HSP90 inhibition**, because aneuploid cells consistently show heightened dependence on chaperone-mediated proteostasis and repeated preclinical sensitivity to HSP90 blockade (Amponsah & Storchová, 2025; Oromendia & Amon, 2014; Kawakami et al., 2019; Simonetti et al., 2018). This fits the biology of SAC failure, where chromosome missegregation creates stoichiometric imbalance in protein complexes, aggregation pressure, increased protein degradation demand, and chronic stress-response dependence (Ben-David & Amon, 2019; Simonetti et al., 2018; Amponsah & Storchová, 2025; Cyran & Zhitkovich, 2022).

The translational picture is less uniform than the mechanistic one. HSP90 inhibitors have the clearest direct link to aneuploid proteotoxic vulnerability, but older pan-HSP90 agents were limited by toxicity and weak monotherapy efficacy in many solid-tumor trials, whereas newer selective agents such as **pimitepib/TAS-116** and biomarker-guided strategies appear more viable (Rastogi et al., 2024; Zarguan et al., 2026; Sanchez et al., 2019). **WEE1 inhibition** has meaningful human oncology evidence, but it targets replication-stress and checkpoint dependence more than proteotoxic stress itself (Di Rorá et al., 2020; Thangaretnam et al., 2025). **Senolytics** are attractive because weakened SAC states and aneuploidy can drive senescence, yet the evidence is mostly indirect, model-specific, and toxicity-limited, especially for navitoclax-class agents (Silva & Bousbaa, 2022; Kirkland & Tchkonja, 2020; Carpenter et al., 2021; Rad & Grillari, 2023).

Are HSP90 inhibitors the most promising repurposing candidates for targeting aneuploidy-induced proteotoxic stress in spindle assembly checkpoint deficiency?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on HSP90 inhibitors for SAC-deficient aneuploidy

2. Methods

This Deep Search synthesis evaluated research on drug repurposing strategies for **aneuploidy-induced proteotoxic stress** in **spindle assembly checkpoint-deficient** disease contexts, especially **BUB1B/BubR1 deficiency**, mosaic variegated aneuploidy, chromosomal instability, and aneuploid cancers. The search ran over more than 220 million research papers in Consensus, spanning Semantic Scholar, PubMed, and other indexed sources.

The workflow identified 2,537 citation-expanded papers in addition to large targeted search returns, screened 230 papers after machine-learned relevance filtering, judged 123 unique papers eligible after deduplication, and included the top 100 most relevant papers for final synthesis. Priority was given to human studies, clinical trials, preclinical cancer models, and mechanistic papers published from 2010-2026 that directly addressed HSP90 inhibitors, WEE1 inhibitors, senolytics, autophagy/proteostasis targeting, or aneuploid stress biology.

Search Strategy

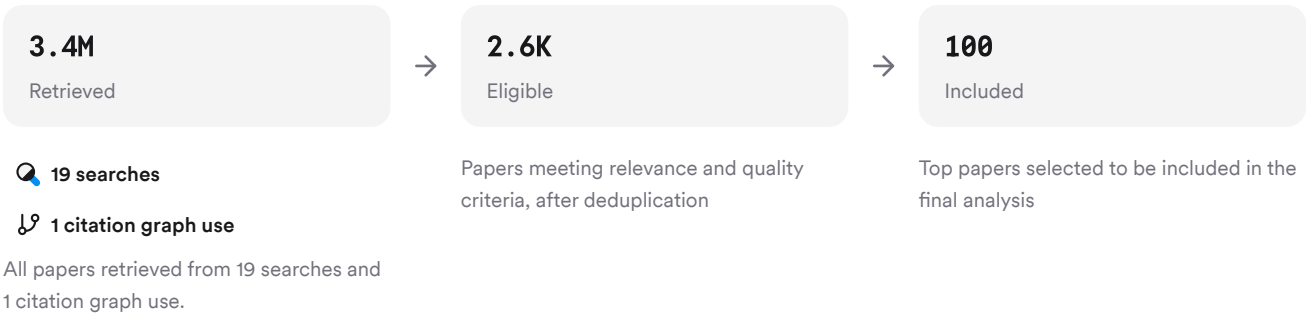


FIGURE 2 Search strategy and paper selection flow

The strategy combined disease-specific, mechanism-specific, and adjacent-pathway searches to connect SAC failure, aneuploidy, proteotoxic stress, and repurposable therapeutics.

3. Results

3.1 Key Papers

A small set of papers anchors the field because they connect **aneuploidy biology** to **druggable proteostasis dependence** and then map which candidate drug classes have actually shown translational traction (Oromendia & Amon, 2014; Kawakami et al., 2019; Rastogi et al., 2024; Schutte et al., 2023). The literature is strongest where mechanistic vulnerability and human oncology experience intersect, which favors **HSP90 inhibitors first** and **WEE1 inhibitors second** (Zarguan et al., 2026; Thangaretnam et al., 2025).

Paper	Summary
(Oromendia & Amon, 2014)	Aneuploid cells show 17-AAG sensitivity (Oromendia & Amon, 2014)
(Kawakami et al., 2019)	17-AAG , chloroquine, AICAR select aneuploid cells (Kawakami et al., 2019)
(Schutte et al., 2023)	Adavosertib has human trial activity (Schutte et al., 2023)

FIGURE 3 Key papers anchoring candidate prioritization

3.2 Candidate Comparison

Dimension	HSP90 Inhibitors	WEE1 Inhibitors	Senolytics	Citations
Direct fit to proteotoxic mechanism	High	Moderate	Moderate	(Rastogi et al., 2024; Ben-David & Amon, 2019; Amponsah & Storchová,

Dimension	HSP90 Inhibitors	WEE1 Inhibitors	Senolytics	Citations
				2025; Di Rorá et al., 2020; Zhang et al., 2022)
Evidence in aneuploid models	Direct	Mostly indirect	Indirect	(Kawakami et al., 2019; Simonetti et al., 2018; Oromendia & Amon, 2014; Cohen-Sharir & Ben-David, 2021; Saleh et al., 2020)
Human oncology experience	Extensive but mixed	Best active program	Early, limited	(Rastogi et al., 2024; Schutte et al., 2023; Kirkland & Tchkonja, 2020)
Main liability	Toxicity, resistance	Myelosuppression, GI toxicity	Thrombocytopenia	(Liang et al., 2024; Sanchez et al., 2019; Thangaretnam et al., 2025; Schutte et al., 2023; Rad & Grillari, 2023; L'hôte et al., 2022)
Best current niche	Biomarker-guided combinations	Replication-stressed tumors	Post-senescence cleanup	(Zarguan et al., 2026; Schutte et al., 2023; Jin et al., 2024)

HSP90 inhibitors are the only queried class with repeated **direct** evidence of selective activity against aneuploid cells. WEE1 inhibitors have stronger contemporary clinical momentum, but through a partially different vulnerability axis.

3.3 HSP90 and WEE1 Evidence

HSP90 inhibition repeatedly emerges as the most mechanistically matched strategy because aneuploid cells are more dependent on folding, degradation, and stress-buffering systems, and multiple reviews converge that **17-AAG** shows enhanced selectivity for aneuploid cells relative to euploid controls (Kawakami et al., 2019; Simonetti et al., 2018; Oromendia & Amon, 2014; Saleh et al., 2020; Ben-David & Amon, 2019). This is especially relevant to mosaic variegated aneuploidy, for which the literature explicitly notes the absence of direct MVA proteotoxicity testing but highlights that complex aneuploid states should be investigated for the same vulnerability (Oromendia & Amon, 2014).

Human HSP90 trials are broad but disappointing as unselected monotherapy: 22 inhibitors across 186 trials produced limited single-agent activity, frequent suspension or termination, and recurrent toxicity signals (Rastogi et al., 2024; Liang et al., 2024; Cools et al., 2025). The more encouraging signals come from **combination therapy**, **proteostasis-burdened tumors**, and newer selective compounds. Multiple myeloma is the clearest example, where combined HSP90 and proteasome inhibition can collapse protein homeostasis and produced clinical responses with tanespimycin plus bortezomib (Zarguan et al., 2026; Trepel et al., 2010; He & Hu, 2018). **TAS-116/pimipib** stands out because it is cytosolic-selective, has less ocular toxicity, has shown partial responses in phase I testing, and reached regulatory approval in refractory GIST, which is the strongest clinical validation of HSP90 targeting in the corpus (Chatterjee & Burns, 2017; Sanchez et al., 2019; Rastogi et al., 2024; Zarguan et al., 2026).

WEE1 inhibition is promising for tumors with **checkpoint addiction**, **TP53 loss**, or **replication stress**, and adavosertib is the best-developed agent (Di Rorá et al., 2020; Van Bijsterveldt et al., 2020; Thangaretnam et al., 2025). Clinical activity is real, especially in gynecologic cancers, where response rates of adavosertib monotherapy or combinations ranged from **23% to 43%** with median PFS **3.0-9.9 months**, but toxicity and biomarker uncertainty remain important constraints (Schutte et al., 2023; Kong & Mehanna, 2021). For rhabdomyosarcoma and other pediatric solid tumors, replication-stress targeting is considered promising, but the strongest pediatric evidence in the corpus favors ATR/CHK1/PARP/WEE1 as a broader class rather than SAC-deficient RMS specifically (Keller et al., 2021).

3.4 Senolytics and Missing IGF-1R Evidence

Senolytics are biologically relevant because weakened SAC function, BubR1 insufficiency, and aneuploidy can drive **premature senescence**, linking SAC deficiency to a senescence-clearance strategy (Silva & Bousbaa, 2022). Within the queried options, **navitoclax/ABT-263** is the best-established senolytic, and **17-DMAG** provides an HSP90-linked senolytic bridge, but senolytic effects are heterogeneous across models and not universal (Kirkland & Tchkonja, 2020; Rad & Grillari, 2023; Carpenter et al., 2021; Saleh et al., 2020).

Feature	Result
Best-established senolytic class	BCL-2/BCL-xL BH3 mimetics (Rad & Grillari, 2023; L'hôte et al., 2022)
Major limitation	Thrombocytopenia from BCL-xL targeting (Rad & Grillari, 2023; L'hôte et al., 2022)
Newer direction	PROTAC or galacto-conjugates to improve selectivity (L'hôte et al., 2022; Liu et al., 2025)
Cancer role	Best suited as one-two-punch adjuncts after senescence induction (Jin et al., 2024; Hamburger et al., 2026)

The corpus provides essentially **no direct support** for **IGF-1R inhibitors** as leading candidates for aneuploidy-induced proteotoxic stress in SAC deficiency. Their absence from the top-ranked evidence, despite broad adjacent-pathway searching, suggests this is presently a weak or underdeveloped strategy in this specific mechanism-defined niche.

Results Timeline

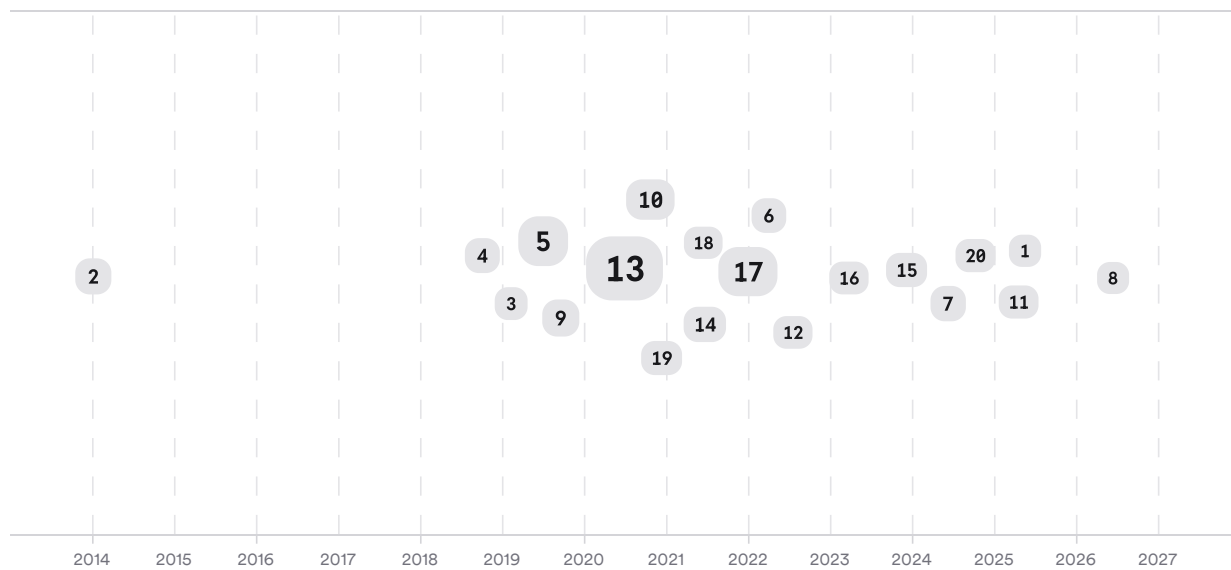


FIGURE 4 Research timeline with larger markers indicating more citations

Top Contributors

Type	Name	Papers
Author	L. Neckers	[7bf800e2ad2750cabf02d9bcbe82b559] [01e337b06c8d5b67b071f7f7a2760034] [697e75fdb1395c9a9ae9b4966ca1f860] [e1af716bdb255ee1a63f3989e6d7d2fe]
Author	G. Chiosis	[7bf800e2ad2750cabf02d9bcbe82b559] [4006e79b4fac57b285d77b86d361011c] [697e75fdb1395c9a9ae9b4966ca1f860]
Author	J. Kirkland	[033e223687c35673893b157517c98907] [da4831ef3e7b5f1699623d2df612cb44]
Journal	<i>International Journal of Molecular Sciences</i>	[7289d9c989725532a4f50d2e4d1c8c2f]
Journal	<i>Cell Stress & Chaperones</i>	[01e337b06c8d5b67b071f7f7a2760034] [c2c9a712f39f5cde891806319650d1b7] [c6c1efae3d215f9ab5df4f3c70b6c9c5]

Type	Name	Papers
Journal	<i>Nature Reviews</i>	[7bf800e2ad2750cabf02d9bcbe82b559] [697e75fdb1395c9a9ae9b4966ca1f860] [d7a4497d0560551e889da0d1d99dccc9]

FIGURE 5 Authors and journals appearing most in included papers

4. Discussion

The literature supports a clear **ranking by mechanistic fit**, not by proven disease-specific efficacy. **HSP90 inhibitors** best match the stated mechanism because they directly exploit the folding and proteostasis burden caused by aneuploidy, and selective killing of aneuploid cells by 17-AAG is one of the few findings reproduced across multiple reviews of the primary screens (Kawakami et al., 2019; Simonetti et al., 2018; Oromendia & Amon, 2014; Saleh et al., 2020). The weakness is translational: most HSP90 trials were not designed around **aneuploidy** or **SAC deficiency** as biomarkers, so negative results in unselected populations do not fully test the specific hypothesis relevant here (Rastogi et al., 2024; Zarguan et al., 2026; Magyar et al., 2023).

WEE1 inhibitors have better present-day human efficacy data than HSP90 inhibitors, but the mechanism is more indirect for proteotoxic stress. They exploit replication stress, checkpoint dependence, and forced mitotic entry, which could be relevant in BUB1B-deficient cancers but does not specifically target the proteostasis burden created by aneuploidy (Di Rorá et al., 2020; Du et al., 2020; Wang et al., 2021). The biomarker story is also incomplete: TP53, CCNE1, RAS, and replication-stress signatures help, but no paper here establishes **BUB1B loss** as a validated predictor of WEE1 response (Schutte et al., 2023; Van Bijsterveldt et al., 2020; Wang et al., 2024).

Senolytics are promising mainly as adjuncts if SAC deficiency produces a substantial senescent compartment in tumor or normal tissues. That logic is plausible from BubR1 aging models and therapy-induced senescence work, but cancer senolysis remains heterogeneous, and navitoclax-class thrombocytopenia is a serious translational barrier (Silva & Bousbaa, 2022; Carpenter et al., 2021; Rad & Grillari, 2023; L'hôte et al., 2022). For personal treatment decisions in a child with MVA syndrome and rhabdomyosarcoma, this evidence base is too indirect to guide therapy without specialist review in pediatric oncology and clinical genetics.

Claim	Evidence Strength	Reasoning	Papers
HSP90 inhibitors are the best mechanistic match for aneuploidy-induced proteotoxic stress	 Strong	Direct aneuploid-selective sensitivity and strong proteostasis rationale	(Oromendia & Amon, 2014; Kawakami et al., 2019; Simonetti et al., 2018; Amponsah & Storchová, 2025)
Newer selective HSP90 agents are more plausible than older pan-HSP90 monotherapy	 Moderate	Clinical field matured, but efficacy remains context-dependent	(Rastogi et al., 2024; Zarguan et al., 2026; Sanchez et al., 2019; Cools et al., 2025)
WEE1 inhibitors are a strong second-line mechanistic option	 Moderate	Good human trial activity, but replication-stress axis is indirect	(Thangaretnam et al., 2025; Schutte et al., 2023; Wang et al., 2024; Keller et al., 2021)

Claim	Evidence Strength	Reasoning	Papers
Senolytics are best viewed as adjunctive, not primary proteotoxic targeting	Moderate	Biological rationale exists, but evidence is model-specific and toxicity-limited	(Silva & Bousbaa, 2022; Carpenter et al., 2021; Rad & Grillari, 2023; Hamburger et al., 2026)
IGF-1R inhibitors are currently leading candidates in this niche	Weak	This corpus provides little direct support	(Amponsah & Storchová, 2025)

FIGURE 6 Key claims and supporting evidence in corpus

5. Conclusion

Across this 2010-2026 corpus, the most defensible answer is that **HSP90 inhibitors** are the leading repurposing candidates for **aneuploidy-induced proteotoxic stress** in **spindle assembly checkpoint deficiency**, with **WEE1 inhibitors** as the strongest clinically active adjacent class rather than the most mechanism-matched one. **Senolytics** are promising mainly for a senescence-clearance strategy, and **IGF-1R inhibitors** lack direct support in this literature.

Research Gaps

The biggest gap is the absence of studies directly testing these agents in **BUB1B-deficient** or **mosaic variegated aneuploidy** models, despite repeated calls to study how complex aneuploid states handle proteotoxic stress in vivo (Oromendia & Amon, 2014; Amponsah & Storchová, 2025).

	Aneuploidy	Clinical trials	Biomarkers	Pediatric models
HSP90	GAP	GAP	GAP	GAP
WEE1	GAP	GAP	GAP	GAP
Senolytics	GAP	GAP	GAP	GAP
Autophagy	GAP	GAP	GAP	GAP

Open Research Questions

Question	Why
Do BUB1B-deficient rhabdomyosarcoma models show greater sensitivity to selective HSP90 inhibitors than euploid controls?	This directly tests the central mechanistic hypothesis missing from current translational evidence.

Question	Why
Can aneuploidy burden or proteostasis biomarkers identify which tumors benefit from HSP90 or WEE1 inhibition?	Most negative trials used unselected populations despite repeated biomarker arguments.
Does sequential senescence induction followed by senolysis improve control of SAC-deficient tumors without excessive toxicity?	Senolytic benefit appears strongest in one-two-punch designs, but safety remains the major barrier.

For this specific SAC-deficient, BUB1B-loss context, the literature most strongly supports **selective HSP90 inhibition first, WEE1 inhibition second, and senolytics as adjuncts rather than primary proteotoxic therapy.**

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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